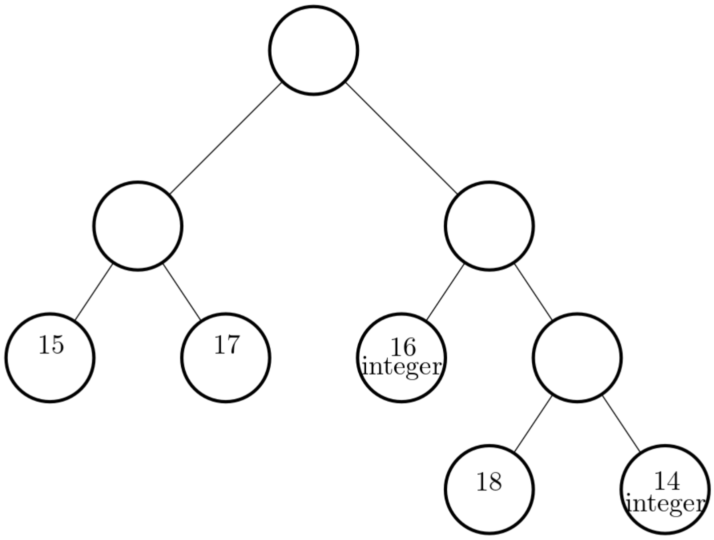
Practice for Quiz 2 IEORE4004, Fall 2025

Instructor: Prof. Yuri Faenza

All exercises from Assignment 4, 5 can be used as practice exercises. Additional exercises are listed below.

1) Branch and Bound

Problem 1: Consider the following Branch and Bound tree for a problem in maximization form. For each leaf node of the current tree, we report the value of the optimal solution to the linear relaxation, and whether the solution is integer.



a) Which are the nodes that can be pruned?

b) Give an upper bound to the value of the optimal integer solution to the original problem.

c) Give a lower bound to the value of the optimal integer solution to the original problem.

d) Now answer a)-b)-c) above assuming the original problem is in minimization form.

Problem 2: Suppose that you are using the Branch-and-bound algorithm to solve the following bounded IP :

$$\max c^T x : \ Ax \leq b, \ x \in \mathbb{Z}^n. \tag{1}$$

For each of the following statements, discuss whether it is true, motivating your answers.

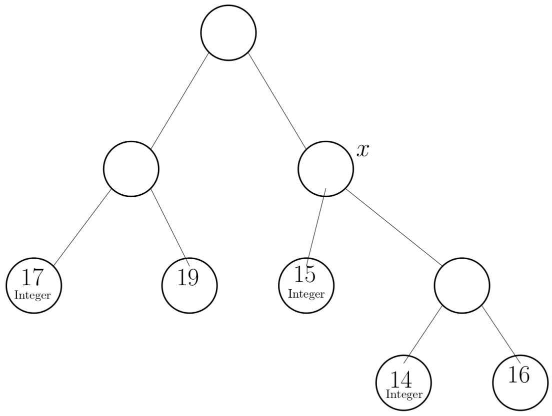
1. If, at some point of the Branch-and-Bound procedure, the optimal integer solution of the current suproblem has value v, the optimal solution of (1) has value at least v.

2. If, at some point of the Branch-and-Bound procedure, the optimal solution of the LR of the current suproblem has value v, the optimal solution of (1) has value at least v.

3. If, at some point of the Branch-and-Bound procedure, the optimal solution to the LR of the current suproblem has value v, the optimal integer solution of each of the two suproblems obtained from the current one by branching will have value strictly less than v

4. If at the end of the Branch-and-Bound procedure we have not found any integer solution, (1) is infeasible.

Problem 3: Consider the following Branch and Bound tree for a problem in maximization form. For each leaf node of the current tree, we report the value of the optimal solution to the linear relaxation, and whether the solution is integer.



a) Which are the nodes that can be pruned? b) Give an upper and a lower bound to the value of the optimal integer solution to the original problem.

c) Now suppose we also have the additional information that in node x, the value of optimal solution to the LP relaxation is 20. How do your answers to $a), b)$ change?

2) Convex Optimization

Problem 4:

a) Let $f, g : \mathbb{R}^d \to \mathbb{R}$ be convex functions. Prove that $f + g$ is a convex function.

b) Prove that $\|x\|$ is convex, where $\|x\|$ is any norm. A norm $\|x\|$ of a vector x is an abstraction of the classical notion of length that satisfy the following properties.

- Triangular inequality: $\|x + y\| \leq \|x\| + \|y\|$
- Homogeneity: For $\lambda \in \mathbb{R}_{\geq 0}$, we have $\|\lambda x\| = \lambda \cdot \|x\|$
- Positiviness: If $\|x\| = 0$, then $x = 0$.

c) Let $f : \mathbb{R}^d \to \mathbb{R}$ be a convex function, and $\alpha \in \mathbb{R}$ be a number. Prove that $C = \{x : f(x) \leq \alpha\}$ is a convex set.

d) Let $\|x\|$ be any norm, and $\alpha \in \mathbb{R}$ be a number. Prove that $C = \{x : \|x\| \leq \alpha\}$ is a convex set.

e) Prove that the portfolio optimization problem seen in class

$$
\begin{array}{lll}
\max & f(x) = 20x_1 + 16x_2 & -\theta[2x_1^2 + x_2^2 + (x_1 + x_2)^2] \\
\text{subject to:} & & \\
& x_1 + x_2 \leq 5 & \\
& x_1 \geq 0 & \\
& x_2 \geq 0 &
\end{array}
$$

can be formulated as a convex programming problem.

Problem 5: Consider the following portfolio optimization problem. An investor has \$10000 and two potential investments. From historical data, investments 1 and 2 have an expected annual return of 13 and 11 percent, respectively. The variance of investing x_1 in investment 1 and x_2 in investment 2 is

$$x_1^2 + 4x_2^2 + 4x_1x_2.$$

a) Formulate as a non-linear optimization problem the problem faced by the investor who wants to maximize the profit subject to a budget constraint and to the fact that the variance is at most α, where α is a given number.

b) Show that the problem from a) can be formulated as a convex programming problem.

c) The *Sharpe ratio* of a portfolio is given by[1]:

$$\frac{\text{expected return}}{\text{standard deviation}}.$$

Show that the problem of maximizing the Sharpe ratio of the portfolio above subject to budget constraints can be formulated as a convex programming problem.

Problem 6: Consider the following problem, faced by a website over the course of T times. There are n advertisers, each of which wishes to show one ad to multiple users. At time t, there are u_t users visiting the website, and the website shows to each of them exactly one of the n ads (the same add can be shown to multiple users). To each advertiser $i = 1, \ldots, n$, the following quantities are associated:

- c_i: this is the minimum number of ads of advertiser i that need to be shown in total, by contract, over the T times;
- $p_{i,t}$: this is the amount an advertiser is willing to pay if their ad is shown to a user that visits the website in time t;
- b_i: this is the maximum budget advertiser i is willing to pay in total over the T times.

a) Define variables

$$x_{i,t} = \text{ number of times add } i \text{ is shown to users arriving at time } t.$$

Describe the amount of money advertiser i will pay as a function of $x_{i,t}$ and other parameters of the problem.

b) Formulate the problem of assigning users to advertisers over the T times as to maximize the revenue of the website ($\equiv$ total expense by the advertisers). You can assume all variables to be fractional.

c) Show that the problem in b) can be formulated as a convex programming problem. You can use the following fact: if $f, g : \mathbb{R}^d \to \mathbb{R}$ are linear functions, then $h : \mathbb{R}^d \to \mathbb{R}$ is a convex function, where $h(x) = \max(f(x), g(x))$ for each $x \in \mathbb{R}^d$.

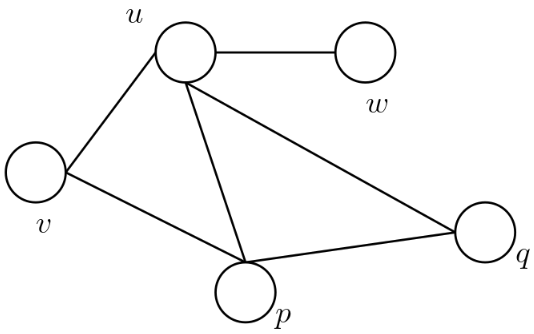
3) IP Modeling

[1]A previous version of this exercise defined the Sharpe ratio as (expected return)/variance, but the most common definition is the one with standard deviation given here.

Problem 7: A *dominating set* of a graph $G(V, E)$ is a set $S \subseteq V$ such that, for each vertex $i \in V$, there exists $j \in S$ such that either $i = j$, or $ij \in E$. The Minimum Dominating Set problem is the following:

- Given: A graph $G(V, E)$;
- Find: A dominating set of G of minimum cardinality.

1. Formulate the Minimum Dominating Set problem for the graph below as an integer program by explicitly listing the objective function and all the constraints.



2. Formulate the Minimum Dominating Set problem for a generic graph as an integer program.

3. Recall that a *vertex cover* of a graph $G(V, E)$ is a set of nodes $S \subseteq V$ such that, for each edge $uv \in E$, $\{u, v\} \cap S \neq \emptyset$. That is, for every edge of the graph, S contains at least one of the endpoints of the edge. Explain why a vertex cover of a graph $G(V, E)$ where each node is adjacent to at least one other node is also a dominating set of $G(V, E)$.

4) Dynamic Programming

Problem 8: There are three possible investments:
Investment 1: investing $x_1 > 0$ dollars yields $c_1(x_1) = 2x_1 + 3$ dollars, while $c_1(0) = 0$.
Investment 2: investing $x_2 > 0$ dollars yields $c_2(x_2) = 3x_2$ dollars, while $c_2(0) = 0$.
Investment 3: investing $x_3 > 0$ dollars yields $c_3(x_3) = x_3^2 + 4$, while $c_3(0) = 0$.
Our total budget is \$4 and we can invest in each investment, either 0 or a multiple of \$1.

Consider the function $f(i, j)$ computing, for each $i = 1, 2, 3$ and $j = 1, 2, \ldots, 4$, the optimal solution of the subproblem where we can only invest in investments $1, 2, \ldots, i$ with a budget of j dollars.

a) Write a recursive formula for the function f. Use the recursive formula to fill in the blanks in the table below that reports the values of $f(i, j)$ (as usual, columns index j, while rows index i). For each of the missing entries, explicitly write the numbers you used to compute it.

	0	1	2	3	4
0	0	0	0	0	0
1	0	5	7	9	11
2	0	5		11	
3	0	5	10		

b) Compute the optimal solution to the problem in a), i.e., how much money we should invest in each investment.

c) Now suppose you can invest multiple of .25$ in each investment (as opposed to multiples of 1$). Write the recursive formula for this new problem (note: you do not need to explicitly compute the table).

Problem 9: There are three possible investments:
Investment 1: investing $x_1 > 0$ dollars yields $c_1(x_1) = 2x_1 + 3$ dollars, while $c_1(0) = 0$.
Investment 2: investing $x_2 > 0$ dollars yields $c_2(x_2) = 3x_2$ dollars, while $c_2(0) = 0$.
Investment 3: investing $x_3 > 0$ dollars yields $c_3(x_3)$, as given by the table below for $x_3 = 1, 2, 3, 4$:

$c_3(1)$	$c_3(2)$	$c_3(3)$	$c_3(4)$
5	8	12	13

while $c_3(0) = 0$. Our total budget is \$4 and, for each investment, we can invest either 0 or a multiple of \$1.

Consider the function $f(i, j)$ computing, for each $i = 1, 2, 3$ and $j = 1, 2, \ldots, 4$, the optimal solution of the subproblem where we can only invest in investments $1, 2, \ldots, i$ with a budget of j dollars.

a) Write a recursive formula for the function f. Use the recursive formula to fill in the blanks in the table below that reports the values of $f(i, j)$ (as usual, columns index j, while rows index i). For each of the missing entries, explicitly write the numbers you used to compute it.

	0	1	2	3	4
0	0	0	0	0	0
1	0	5	7	9	11
2	0	5		11	
3	0	5	10		

b) Compute the optimal solution to the problem in a), i.e., how much money we should invest in each investment.

c) After a change in the market, the yield of investment 2 decreases substantially: now investing x_2 dollars yields $c_2(x_2) = 1.5x_2$ dollars. Without computing the table again, argue why the optimal solution to the problem does not change.

Problem 10: We have 2 investments giving a return of 11 and 14 percent, respectively. There is moreover some risk associated with those investments, and the variance of the expected return is $4x_1^2 + 9x_2^2 + 6x_1x_2 + 1$, when x_1 dollars are invested in the first investment, and x_2 dollars are invested in the second investment. Our budget is \$10.

a) Formulate the problem of finding a portfolio that, among those that respect the budget, maximizes the expected return minus the variance. Explain why it is a convex program by analyzing its objective function and constraints, and performing the appropriate calculations, if needed.

b) Formulate the problem of finding a portfolio that, among those that respect the budget and whose expected return is at least a given value α, minimizes the variance. Explain why it is a convex program by analyzing its objective function and constraints, and performing the appropriate calculations, if needed.

c) We now want to modify the portfolio in b) so that we also limit our total exposure. We therefore add to the formulation in part b) the constraint that the variance plus $\|x\|_2$ should not exceed a given value γ. Is the problem still a convex program? Explain.

Problem 11:

a) You are given a 4×7 board and a robot initially placed at the upper left corner. At each step, the robot can either move one cell to the right, or one cell down. Assume that traps are placed in some cells. Write a dynamic program to help you decide if the robot can arrive from the origin to the destination without passing through a trap.

(b) Suppose now that the robot can, at each step, perform one of the following moves: move one position to the right; move two positions to the right, jumping over one cell (and the trap on that cell, if any); move one position down; move two position down, jumping over one cell (and the trap on that cell, if any). Write a dynamic program to help you decide if the robot can go from the origin to the destination using these moves and, if it can, find the smallest number of moves that brings the robot from the origin to the destination.

For each of the bullet points above, explicitly report the recursive function, the table, and the optimal solution.

Bonus question #1. Recall the scheduling problem seen in class. We have n jobs $j_1, j_2, \ldots, j_n$ that need to be completed. Any job can be scheduled on any of 2 available machines, but no two jobs can be scheduled on the same machine in overlapping times. The processing times $p_1, p_2, \ldots, p_n$ do not depend on the machine they are scheduled on. The goal is to schedule jobs on machines as to minimize the time when all jobs have been completed.

Formulate a dynamic program that solves the modification of the problem above where there are 3 available machines.

Bonus question #2. Consider a knapsack instance with n items, profits $p_1, \ldots, p_n \in \mathbb{N}$, capacity β, and weights $w_1, \ldots, w_n \in \mathbb{N}$. Give an efficient algorithm that computes all items that are in some optimal solution of a knapsack problem.